Student Performance Analysis System in Java

Aim / Objective

To design and implement a student performance analysis system in Java using Object-Oriented Programming principles to calculate marks, percentage, grade, highest and lowest marks, and overall academic performance.

Description

The Student Performance Analysis System accepts a student's roll number, name, and five subject marks. It uses a class and object to calculate total, average, highest and lowest marks, percentage, grade, result, and a simple performance remark. String operations are used to format the student name and Math methods are used for percentage calculation.

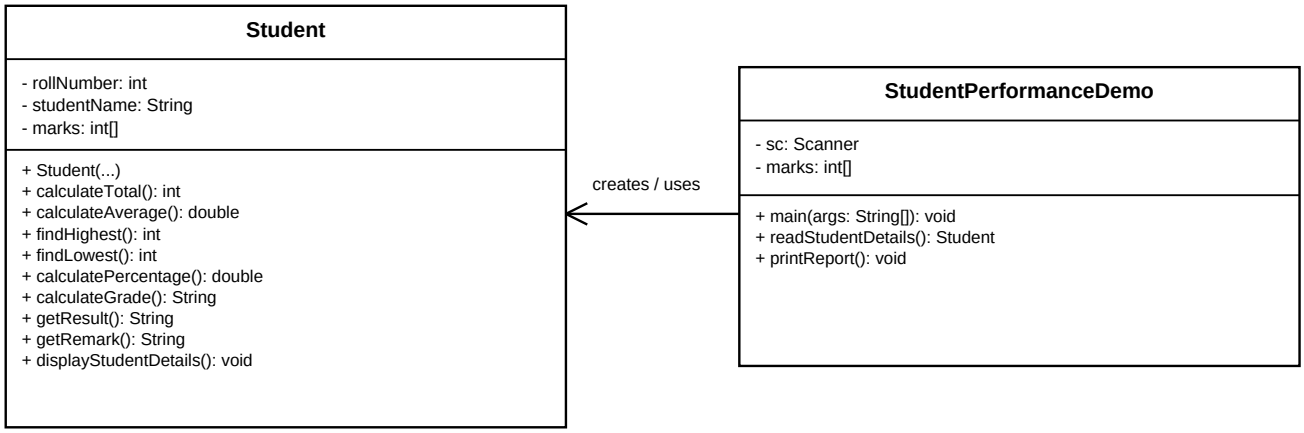
System Requirements

Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.
Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm

1. Start the program.
2. Create a Scanner object for user input.
3. Read the roll number and student name.
4. Create an integer array to store marks of 5 subjects.
5. Read the marks of all 5 subjects using a loop.
6. Create a Student object using the entered details.
7. Calculate the total marks by adding all array elements.
8. Calculate the average and percentage from the total marks.
9. Find the highest and lowest marks using Math.max() and Math.min().
10. Round the percentage to two decimal places using Math.round().
11. Format the student name using trim(), toUpperCase(), and length().
12. Assign the grade according to the percentage.
13. Determine Pass or Fail based on a percentage of 50% or above.
14. Display the complete performance report and close Scanner.

UML Diagram



Source Code

```
import java.util.Scanner;

class Student {
    private int rollNumber;
    private String studentName;
    private int[] marks;

    // Constructor
    Student(int rollNumber, String studentName, int[] marks) {
        this.rollNumber = rollNumber;
        this.studentName = studentName;
        this.marks = marks;
    }

    // Calculate total marks
    int calculateTotal() {
        int total = 0;

        for (int mark : marks) {
            total += mark;
        }

        return total;
    }

    // Calculate average marks
    double calculateAverage() {
        return calculateTotal() / 5.0;
    }
```

```
    int highest = marks[0];

    for (int mark : marks) {
        highest = Math.max(highest, mark);
    }

    return highest;
}

// Find lowest mark
int findLowest() {
    int lowest = marks[0];

    for (int mark : marks) {
        lowest = Math.min(lowest, mark);
    }

    return lowest;
}

// Calculate percentage
double calculatePercentage() {
    double percentage = (calculateTotal() / 500.0) * 100;

    return Math.round(percentage * 100.0) / 100.0;
}

// Calculate grade
String calculateGrade() {
    double percentage = calculatePercentage();

    if (percentage >= 90)
        return "A+";
    else if (percentage >= 80)
        return "A";
    else if (percentage >= 70)
        return "B";
    else if (percentage >= 60)
        return "C";
    else if (percentage >= 50)
        return "D";
    else
        return "F";
}

// Check pass or fail
String getResult() {
    if (calculatePercentage() >= 50)
        return "Pass";
    else
        return "Fail";
}

// Performance remark
String getRemark() {
    String grade = calculateGrade();

    if (grade.equals("A+"))
        return "Excellent Performance";
    else if (grade.equals("A"))
        return "Very Good Performance";
    else if (grade.equals("B"))
        return "Good Performance";
```

```
        else if (grade.equals("D"))
            return "Needs Improvement";
        else
            return "Poor Performance";
    }

    // Display student details
    void displayStudentDetails() {
        String formattedName = studentName.trim().toUpperCase();

        System.out.println("\n===== PERFORMANCE REPORT =====");
        System.out.println("Roll Number       : " + rollNumber);
        System.out.println("Student Name      : " + formattedName);
        System.out.println("Name Length       : " + formattedName.length());
        System.out.println("Total Marks       : " + calculateTotal());
        System.out.println("Average Marks     : " + calculateAverage());
        System.out.println("Highest Mark      : " + findHighest());
        System.out.println("Lowest Mark       : " + findLowest());
        System.out.printf("Percentage        : %.2f%%%n", calculatePercentage());
        System.out.println("Grade             : " + calculateGrade());
        System.out.println("Result            : " + getResult());
        System.out.println("Performance Remark: " + getRemark());
    }
}

public class StudentPerformanceDemo {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("===== STUDENT PERFORMANCE ANALYSIS =====");

        System.out.print("Enter Roll Number: ");
        int rollNumber = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Student Name: ");
        String studentName = sc.nextLine();

        int[] marks = new int[5];

        System.out.println("\nEnter marks for 5 subjects:");

        for (int i = 0; i < 5; i++) {
            System.out.print("Enter marks for Subject " + (i + 1) + ": ");
            marks[i] = sc.nextInt();
        }

        Student student = new Student(rollNumber, studentName, marks);

        student.displayStudentDetails();

        sc.close();
    }
}
```

Compilation and Execution Commands

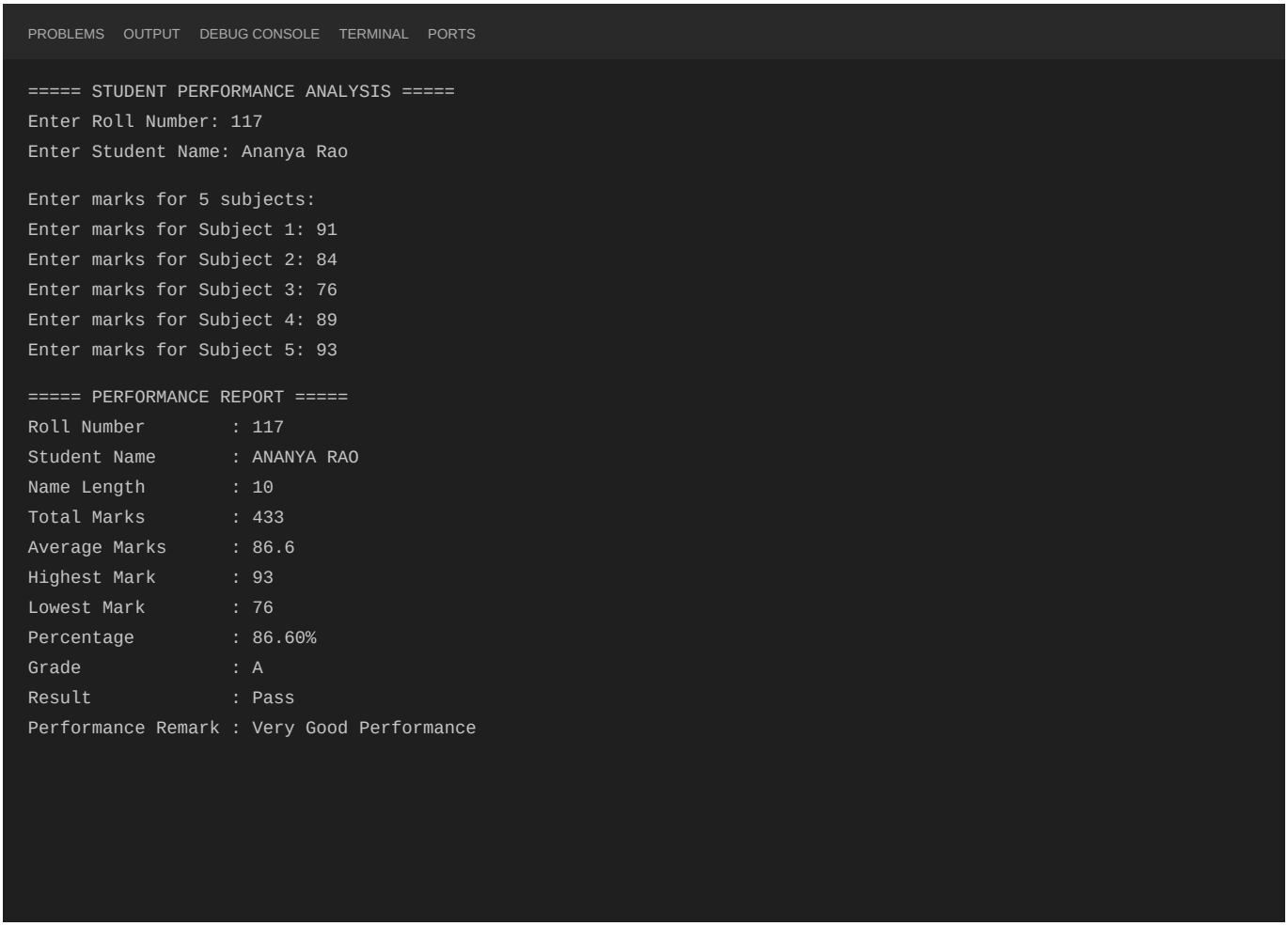
Save as: StudentPerformanceDemo.java

Compile: javac StudentPerformanceDemo.java

Execute: java StudentPerformanceDemo

Output Screens

Possible Output – Successful Performance Analysis



Possible Output – Failed Performance Analysis

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
===== STUDENT PERFORMANCE ANALYSIS =====
Enter Roll Number: 124
Enter Student Name: Kiran Das

Enter marks for 5 subjects:
Enter marks for Subject 1: 42
Enter marks for Subject 2: 38
Enter marks for Subject 3: 49
Enter marks for Subject 4: 45
Enter marks for Subject 5: 41

===== PERFORMANCE REPORT =====
Roll Number        : 124
Student Name       : KIRAN DAS
Name Length        : 9
Total Marks        : 215
Average Marks      : 43.0
Highest Mark       : 49
Lowest Mark        : 38
Percentage         : 43.00%
Grade              : F
Result             : Fail
Performance Remark : Poor Performance
```

Result

Thus, the Student Performance Analysis System was implemented successfully in Java using Object-Oriented Programming concepts, arrays, String operations, Math methods, loops, and conditional statements. The program successfully calculates and displays the student's total marks, average, highest and lowest marks, percentage, grade, result, and performance remark.